Science
Lesson Plan
Grade 10

Teacher Name	Topic:	No of periods required:
Aayushi Kasera	**Electric Circuits**	**3**

Period No	Concept	Learning Objective	Learning Outcomes	Teacher-Learning Process	Assessment	Resources	21st Century Skills/Value Education	Real Life Application	Reflection
1	This period introduces the fundamental concepts of electric current, potential difference, and resistance, which are the building blocks for understanding electric circuits. Students will grasp how these quantities are defined and measured, setting the stage for Ohm's Law.	Students will learn: • To define electric current, potential difference, and resistance. • To understand the units of measurement for each quantity (Ampere, Volt, Ohm). • To identify the components of a simple electric circuit. • To differentiate between conductors and insulators.	Students will be able to: • Define and recall the SI units of electric current, potential difference, and resistance (Remembering). • Explain the flow of charge in a conductor (Understanding). • Construct a simple circuit diagram using standard symbols (Applying). • Compare and contrast conductors and insulators with examples (Analyzing).	The teacher will engage students by asking about daily uses of electricity, explore through a simple circuit demonstration, explain definitions and units using visual aids and analogies, elaborate with a hands-on activity of identifying circuit components, and evaluate understanding through a quick quiz.	• Oral questions: 'What is the unit of potential difference?' (Remembering) • Diagram labeling: Students label components of a given circuit diagram (Understanding). • Short answer questions: 'Explain the role of a switch in an electric circuit.' (Understanding) • Activity observation: Assess student participation and understanding during the circuit component identification activity (Applying).	• Whiteboard and markers • Textbooks (NCERT Class 10 Science) • Projector and presentation slides • Simple circuit kit (battery, wires, bulb, switch, ammeter, voltmeter) • Examples of conductors and insulators (e.g., copper wire, plastic, wood)	• Critical Thinking: Analyzing the function of different circuit components. • Communication: Explaining concepts clearly. • Collaboration: Working in groups for circuit activities. • Scientific Temper: Developing a logical approach to understanding electrical phenomena. • Value of safety: Understanding the importance of safe handling of electrical appliances.	• Understanding how household appliances like lights and fans work. • Identifying the importance of proper wiring and insulation in homes. • Recognizing the role of batteries in portable electronic devices.	• What are the three main quantities we discussed today, and how are they related? • Can you think of a situation where understanding electric current could be important for safety? • How does the flow of water in a pipe relate to the flow of electric current?
2	This period delves into Ohm's Law, a fundamental principle linking potential difference, current, and resistance. Students will understand its mathematical formulation, graphical representation, and the concept of resistance, including factors affecting it.	Students will learn: • To state and explain Ohm's Law. • To understand the relationship between V, I, and R. • To plot V-I graphs for a metallic conductor. • To identify factors affecting the resistance of a conductor (length, area of cross-section, material).	Students will be able to: • State Ohm's Law and its mathematical expression (Remembering). • Interpret V-I graphs to determine resistance (Understanding). • Calculate resistance, current, or potential difference using Ohm's Law (Applying). • Analyze how changes in length, area, or material affect a conductor's resistance (Analyzing).	The teacher will engage students with a scenario involving varying brightness of a bulb, explore Ohm's Law through a virtual lab simulation or a physical experiment, explain the law and its implications, elaborate by solving numerical problems and discussing factors affecting resistance, and evaluate through a quiz on Ohm's Law and problem-solving.	• Quiz: Multiple-choice questions on Ohm's Law and its formula (Remembering, Understanding). • Problem Solving: Numerical problems involving V, I, R calculations (Applying). • Graph Analysis: Students interpret a given V-I graph to find resistance (Analyzing). • Activity: Students	• Whiteboard and markers • Textbooks (NCERT Class 10 Science) • Projector and presentation slides • Online simulation for Ohm's Law (e.g., PhET simulations) • Materials for a simple Ohm's Law experiment (resistors, ammeter, voltmeter, battery, rheostat, wires) • Mind Map template for Ohm's Law.	• Problem Solving: Applying Ohm's Law to solve numerical problems. • Data Interpretation: Analyzing V-I graphs. • Scientific Inquiry: Designing and conducting simple experiments. • Accuracy: Emphasizing precision in measurements and calculations. • Ethical Use of Technology: Understanding the proper use of electrical devices.	• Designing circuits for specific current or voltage requirements. • Understanding why different appliances have different resistances (e.g., heater vs. light bulb). • Troubleshooting common electrical issues by applying Ohm's Law.	• How does Ohm's Law help us predict the current in a circuit? • Why do some electrical wires get hot when current flows through them? • If you wanted to reduce the current in a circuit without changing the voltage, what would you do to the resistance?

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			• Design a simple experiment to verify Ohm's Law (Creating).		plot a V-I graph from given data and explain their findings (Applying, Analyzing).				
3	This period focuses on the concept of resistance and its dependence on various factors. Students will explore the resistivity of materials and understand how it influences the choice of materials for different electrical applications.	Students will learn: • To define resistance and resistivity. • To understand the factors affecting the resistance of a conductor (length, area of cross-section, nature of material). • To differentiate between good conductors, resistors, and insulators based on resistivity values. • To solve numerical problems related to resistivity.	Students will be able to: • Define resistivity and state its SI unit (Remembering). • Explain the relationship between resistance, length, area, and resistivity (Understanding). • Calculate resistance or resistivity given relevant parameters (Applying). • Compare the resistivity of different materials and justify their uses (Analyzing). • Create a mind map summarizing the factors affecting resistance (Creating).	The teacher will engage students by asking why some wires are thicker than others, explore through a discussion on different materials' electrical properties, explain the concept of resistivity and its formula, elaborate by solving numerical problems and discussing practical applications, and evaluate through a quiz and a mind map creation activity.	• Quiz: Short answer questions on definitions of resistance and resistivity (Remembering). • Numerical Problems: Calculate resistance or resistivity given specific values (Applying). • Comparative Analysis: 'Why is copper used for electrical wiring and nichrome for heating elements?' (Analyzing). • Mind Map Creation: Students create a mind map on 'Factors Affecting Resistance' (Creating). • Activity: Group discussion on selecting materials for specific electrical applications (Evaluating).	• Whiteboard and markers • Textbooks (NCERT Class 10 Science) • Projector and presentation slides • Chart showing resistivity values of common materials • Sample wires of different materials and thicknesses • Mind Map software/templates or chart paper for mind map activity.	• Critical Thinking: Analyzing material properties for specific applications. • Problem Solving: Applying formulas to calculate resistance and resistivity. • Creativity: Designing mind maps to organize information. • Environmental Awareness: Understanding the sustainable use of resources in electrical engineering. • Innovation: Thinking about how material science impacts technology.	• Choosing appropriate wires for household wiring based on current capacity. • Understanding why heating elements are made of alloys like nichrome. • Designing components in electronic devices that require specific resistance values. • Importance of material selection in manufacturing electrical cables.	• How does the thickness of a wire affect its resistance, and why is this important in practical applications? • If you had to design a new electrical appliance, what factors related to resistance and resistivity would you consider? • What are some examples of materials with very high resistivity, and where are they used?